

Generation Interconnection System Impact Study Report

For Queue Project Q18-1076

Lick 100 kV
August 24, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1076. The Interconnection Customer proposes a 254.8 MW natural gas combined cycle facility interconnecting at the Lick 100 kV substation in the Ohio area. The request was received on October 12, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 457 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$60,065,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The SONORA – LICK 100 kV is overloaded to 156.2% of its 196 MVA emergency rating for the loss of the SUN VLY – MAGNOLIA RANCH N 100 kV. The project contributes 49.5% DFAX (126.1 MW) to this overload.

For the loss of WEST LAFAYETTE – MOSSYROCK 100 kV, the ANDREWS – SONORA 100 kV L loads to 144.9% of its 135 MVA rating. Project DFAX: 19.8%.

For the loss of WEST LAFAYETTE – MOSSYROCK 100 kV, the OAKVILLE – ANDREWS 100 kV loads to 142.4% of its 129 MVA rating. Project DFAX: 19.8%.

For the loss of MOSSYROCK – W LAFAYETTE 100 kV line, the OAKVILLE – JUNEAU 100 kV loads to 138.3% of its 124 MVA rating. Project DFAX: 19.8%.

The HARTLAND 138-1 kV Transformer is overloaded to 136.1% of its 198 MVA emergency rating for the loss of the HUBBARDSTOWN – TRISTATE 345 kV. The project contributes 5.2% DFAX (13.3 MW) to this overload.

The SUN VALLEY – NORSEAL SUBSTATION 100 kV is overloaded to 131.2% of its 168 MVA emergency rating for the loss of the SONORA – LICK 100 kV. The project contributes 37.7% DFAX (96.1 MW) to this overload.

MAGNOLIA RANCH NORTH – LICK 100 kV: 129.9% loading (196 MVA) following the outage of the SONORA – LICK 100 kV; DFAX = 100.0%.

The SUN VLY – MAGNOLIA RANCH N 100 kV Line is overloaded to 128.2% of its 180 MVA emergency rating for the loss of the LICK – SONORA 100 kV. The project contributes 37.7% DFAX (96.1 MW) to this overload.

For the loss of TIDD – MUSKINGUM RIVR 500 kV L, the EL PASO – RAVENNA 500 kV Line loads to 124.2% of its 2,948 MVA rating. Project DFAX: 10.2%.

HYLEBOS – RAVENNA 500 kV: 120.2% loading (2,504 MVA) following the outage of the TIDD – MUSKINGUM RIV 500 kV; DFAX = 6.3%.

The MCLOUGHLIN – HARTLAND 138 kV is overloaded to 120.1% of its 295 MVA emergency rating for the loss of the HBBARDSTOWN – TRISTATE 345 kV. The project contributes 5.7% DFAX (14.6 MW) to this overload.

RAVENNA 230-1 kV TX: 115.4% loading (1,411 MVA) following the outage of the HYLEBOS SUBSTATION – RAVENNA 500 kV; DFAX = 5.2%.

For the loss of HYLEBOS – RAVENNA 500 kV, the RAVENNA 500/1 kV Transformer loads to 114.0% of its 1,428 MVA rating. Project DFAX: 5.2%.

The NORWICH – BARRETT 500 kV is overloaded to 112.1% of its 3,374 MVA emergency rating for the loss of the SWAN VLY – TREGO 500 kV L. The project contributes 7.3% DFAX (18.6 MW) to this overload.

GAVIN POWER LLC – OLYMPUS 230 kV Line: 112.0% loading (1,231 MVA) following the outage of the GAVIN POWER LLC 500/345 kV Autotransformer; DFAX = 10.4%.

For the loss of MIGUEL – THUNDER 100 kV Ckt 2, the THUNDER – MIGUEL 100 kV Circuit 1 loads to 111.3% of its 156 MVA rating. Project DFAX: 39.7%.

The THUNDER – MIGUEL 100 kV #2 is overloaded to 111.3% of its 156 MVA emergency rating for the loss of the MIGUEL SUBSTATION – THUNDER 100 kV. The project contributes 39.7% DFAX (101.3 MW) to this overload.

For the loss of PINEY CRK PROJECT – HILLTOP SUBSTATION 500 kV, the FORT GIBSON – ARMSTRONG 500 kV L loads to 110.7% of its 2,513 MVA rating. Project DFAX: 8.5%.

For the loss of RAVENNA – EL PASO 500 kV, the TIDD – MSKINGUM RIVER 500 kV line loads to 110.2% of its 2,844 MVA rating. Project DFAX: 11.4%.

GAVIN POWER LLC SUBSTATION 500/345 kV Transformer: 110.0% loading (1,310 MVA) following the outage of the GAVIN POWER LLC – OLYMPUS 230 kV; DFAX = 7.6%.

SIMPSON STUART TAP – MEDORA 345 kV: 109.9% loading (1,452 MVA) following the outage of the HAMPTONRD – MEDORA 345 kV; DFAX = 5.5%.

CORONET – WESTWOOD GENERATION LLC 500 kV: 107.7% loading (3,533 MVA) following the outage of the ACADIA – MONTOUR STA 500 kV Line; DFAX = 8.8%.

The ORCHARD RD – FORT HUMBUG 500 kV L is overloaded to 107.4% of its 4,302 MVA emergency rating for the loss of the W CAMPUS – FT HUMBUG 345 kV. The project contributes 10.4% DFAX (26.5 MW) to this overload.

The ALLEGHENY ENERGY UNITS 3 4 & 5 – SUGAR HILL 345 kV is overloaded to 106.1% of its 2,356 MVA emergency rating for the loss of the HILLTOP SUBSTATION – PINEY CREEK PROJECT 500 kV. The project contributes 8.2% DFAX (20.8 MW) to this overload.

The ALLEGHENY ENERGY UNITS 3 4 & 5 345-1 kV Autotransformer is overloaded to 106.0% of its 2,177 MVA emergency rating for the loss of the HILLTOP – PINEY CRK PROJECT 500 kV line. The project contributes 8.1% DFAX (20.5 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
SONORA – LICK 100 kV Line	Rebuild line with higher-capacity conductor	\$3,386,000	53% (\$1,795,000)
ANDREWS – SONORA SUBSTATION 100 kV	Rebuild line with higher-capacity conductor	\$7,551,000	53% (\$4,009,000)
OAKVILLE – ANDREWS 100 kV	Rebuild line with higher-capacity conductor	\$16,719,000	53% (\$8,875,000)
OAKVILLE – JUNEAU 100 kV	Rebuild line with higher-capacity conductor	\$6,510,000	53% (\$3,456,000)
HARTLAND 138-1 kV Transformer	Replace transformer with higher-rated unit	\$7,477,000	17% (\$1,283,000)
SUN VALLEY – NORSEAL 100 kV	Rebuild line with higher-capacity conductor	\$7,971,000	50% (\$3,999,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
MAGNOLIA RANCH NORTH – LICK 100 kV	Rebuild line with higher-capacity conductor	\$2,170,000	100% (\$2,170,000)
SUN VLY – MAGNOLIA RANCH N 100 kV Line	Rebuild line with higher-capacity conductor	\$3,821,000	50% (\$1,917,000)
EL PASO – RAVENNA 500 kV Line	Rebuild line with higher-capacity conductor	\$210,083,000	2% (\$5,010,000)
HYLEBOS – RAVENNA 500 kV	Rebuild line with higher-capacity conductor	\$119,696,000	2% (\$2,936,000)
HARTLAND – MCLOUGHLIN 138 kV L	Rebuild line with higher-capacity conductor	\$3,447,000	15% (\$503,000)
RAVENNA 230-1 kV TX	Replace transformer with higher-rated unit	\$10,775,000	3% (\$310,000)
RAVENNA 500-1 kV TX	Replace transformer with higher-rated unit	\$36,762,000	3% (\$1,057,000)
NORWICH – BARRETT 500 kV	Rebuild line with higher-capacity conductor	\$63,936,000	1% (\$787,000)
GAVIN POWER LLC – OLYMPUS 230 kV line	Rebuild line with higher-capacity conductor	\$7,687,000	4% (\$275,000)
THUNDER – MIGUEL 100 kV Circuit 1	Rebuild line with higher-capacity conductor	\$4,581,000	36% (\$1,656,000)
MIGUEL – THUNDER 100 kV Line 2	Rebuild line with higher-capacity conductor	\$12,887,000	36% (\$4,660,000)
FORT GIBSON – ARMSTRONG 500 kV L	Rebuild line with higher-capacity conductor	\$97,457,000	2% (\$1,916,000)
TIDD – MSKINGUM RIVER 500 kV line	Rebuild line with higher-capacity conductor	\$435,435,000	2% (\$10,434,000)
GAVIN POWER LLC SUBSTATION 500/345 kV Transformer	Replace transformer with higher-rated unit	\$26,431,000	3% (\$803,000)
MEDORA – SIMPSON STUART TAP 345 kV	Reconductor line	\$21,359,000	2% (\$456,000)
CORONET – WESTWOOD GENERATION LLC 500 kV	Reconductor line	\$49,155,000	2% (\$945,000)
ORCHARD RD – FT HUMBUG 500 kV line	Reconductor line	\$18,514,000	2% (\$360,000)
ALLEGHENY ENERGY UNITS 3 4 & 5 – SUGAR HILL 345 kV	Reconductor line	\$5,052,000	2% (\$105,000)
ALLEGHENY ENERGY UNITS 3 4 & 5 345-1 kV Autotransformer	Replace transformer with higher-rated unit	\$16,912,000	2% (\$348,000)
	Total allocated to project		\$60,065,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.